Circles Maze ANSWER KEY

Start top-left. Leave answers in terms of π and in simplest radical form. Go through the door with your answer.

START

1 Find the circumference.

$r = 6$

12π

2 Find the area.

$d = 10$

5π

Find the radius.

$(x - 6)^2 + y^2 = 81$

81

Find the area.

circumference 16π

24π

146°

100π

25π

220°

125°

256π

Triangle ABC is inscribed in a circle. AB is a diameter.

$m\angle A = 34°$, $m\angle B = ?$

34°

6 Find the sector area.

$r = 6$

central angle 90°

18π

3 Find the inscribed angle.

intercepted arc 110°

110°

Find the central angle that intercepts the same arc.

inscribed angle 40°

(3, 5)

9π

3π

36π

55°

40°

20°

7 Find the center.

$(x - 3)^2 + (y + 5)^2 = 49$

(−3, 5)

5 Find the arc length.

$r = 9$

central angle 60°

70°

4 Find the intercepted arc.

inscribed angle 35°

145°

Find the center.

$(x + 1)^2 + (y - 2)^2 = 16$

(3, −5)

20

$\frac{27\pi}{2}$

6π

35°

17.5°

(1, −2)

8 Find the radius.

$x^2 + (y - 4)^2 = 20$

$2\sqrt{5}$

9 Find the arc length.

$r = 12$

central angle 150°

10π

10 Find the sector area.

$r = 10$

central angle 72°

20π

FINISH

You made it!

Circles Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

1	Answer: **12π**	$C = 2\pi r \rightarrow C = 2\pi(6) = 12\pi$
2	Answer: **25π**	$r = 5 \rightarrow A = \pi r^2 = 25\pi$
3	Answer: **55°**	$\text{angle} = \frac{1}{2}(\text{arc}) \rightarrow \frac{1}{2}(110°) = 55°$
4	Answer: **70°**	$\text{arc} = 2(\text{angle}) \rightarrow 2(35°) = 70°$
5	Answer: **3π**	$L = \frac{60}{360} \cdot 2\pi(9) \rightarrow L = \frac{1}{6}(18\pi) = 3\pi$
6	Answer: **9π**	$A = \frac{90}{360} \cdot \pi(6)^2 \rightarrow A = \frac{1}{4}(36\pi) = 9\pi$
7	Answer: **(3, −5)**	$(x-h)^2 + (y-k)^2 = r^2 \rightarrow h = 3, k = -5$
8	Answer: **$2\sqrt{5}$**	$r^2 = 20 \rightarrow r = \sqrt{20} = 2\sqrt{5}$
9	Answer: **10π**	$L = \frac{150}{360} \cdot 2\pi(12) \rightarrow L = \frac{5}{12}(24\pi) = 10\pi$
10	Answer: **20π**	$A = \frac{72}{360} \cdot \pi(10)^2 \rightarrow A = \frac{1}{5}(100\pi) = 20\pi$